

Activity

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1st Sequence

$$\text{Push}(54) = [54]$$

$$\text{Push}(52) = [54, 52]$$

$$\text{Pop}() = [54]$$

$$\text{Push}(55) = [54, 55]$$

$$\text{Push}(62) = [54, 55, 62]$$

$$S = \text{pop}() = 62$$

2nd Sequence

$$\text{enqueue}(21) = [21]$$

$$\text{enqueue}(24) = [21, 24]$$

$$\text{dequeue}() = [24]$$

$$\text{enqueue}(28) = [24, 28]$$

$$\text{enqueue}(32) = [24, 28, 32]$$

$$q = \text{dequeue}();$$

$$q = 24$$

$$S + q = 62 + 24$$

$$= 86$$

$$2. \quad 10 \ 5 \ + \ 60 \ 6 / \ * \ 8$$

$$[10, 5]$$

$$[15]$$

$$[15, 60]$$

$$[15, 60, 6]$$

$$[15, 10]$$

$$[150]$$

$$[150, 8]$$

$$[150, 8]$$

$$= 142$$

$$3. \quad 8 \ 2 \cdot 3 \wedge 1 \cdot 2 \cdot 3 \cdot + \ 5 \wedge 1 \cdot -$$

$$[8, 2, 3]$$

$$[8, 2^3] = [8, 8]$$

$$= 1$$

$$= [1, 2, 3]$$

$$= [1, 6]$$

Top two elements of stack
after first multiplication

$$is [6, 1]$$

$$4) a + b + c - d \wedge e \wedge f$$

Variable	Stack	Output
a	+	a
+	+	a
b	+ *	ab.
x	+ *	abc
c	+ -	abc * +
-	+ - ^	abc * + d
d	+ - ^	abc * + d
^	* - ^	abc * + d
e	+ - *	abc * + de
^	- ^ ^	abc * + de
f	- ^ ^ +	abc * + de f

$$abc * + de f \wedge \wedge$$

$$5) \text{push}(10) = [10]$$

$$\text{" (20)} : [10, 20]$$

$$\text{pop}() = [10]$$

$$\text{push}(10) = [10, 10]$$

$$\text{push}(20) = [10, 10, 20]$$

$$\text{pop} = [10, 10]$$

$$\text{pop} = [10]$$

$$\text{pop} = []$$

$$\text{push}(20) = [20]$$

$$\text{pop} = []$$

Sequence of

Pop: [20, 20,

10, 20]